

Lipschitz Continuity

A concise overview for graduate students familiar with real analysis.

Definition

Let (X, d_X) and (Y, d_Y) be metric spaces.

A map $f : X \rightarrow Y$ is Lipschitz if there exists $L \geq 0$ such that

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

- Such an L is a Lipschitz constant; global vs. local Lipschitz as usual.

Geometric intuition

- On $\mathbb{R}$, $|f(x) - f(y)| \leq L|x - y|$ means every chord has slope (in absolute value) at most L .
- Picture: no secant of the graph is steeper than L — a uniform slope bound.

Example 1: linear maps

- $f(x) = ax + b$ on $\mathbb{R}$: $|f(x) - f(y)| = |a| |x - y|$.
- Hence Lipschitz with constant $L = |a|$ (minimal).

Example 2: bounded derivative

- If $f \in C^1([a, b])$ and $L = \sup_{[a, b]} |f'| < \infty$, then for all $x, y \in [a, b]$,

$$|f(x) - f(y)| \leq L|x - y|.$$

- Therefore, f is Lipschitz with constant L .

A continuous non-Lipschitz example

- $f(x) = \sqrt{x}$ on $[0, 1]$ is continuous but not Lipschitz on any interval containing 0.
- Indeed $\frac{|\sqrt{x} - \sqrt{0}|}{|x - 0|} = \frac{1}{\sqrt{x}} \rightarrow \infty$ as $x \downarrow 0$.

Uniform continuity & short MVT proof

- Every Lipschitz map is uniformly continuous: given $\varepsilon > 0$ take $\delta = \varepsilon/L$.
- If $f \in C^1([a, b])$ and $L = \sup_{[a, b]} |f'| < \infty$, then for any $x, y \in [a, b]$, there exists ξ between x and y such that $f(x) - f(y) = f'(\xi)(x - y)$.
- Therefore, $|f(x) - f(y)| \leq L|x - y|$.

Takeaway

- Lipschitz = uniform linear control on increments with constant L .
- Implies uniform continuity and provides a simple, uniform slope bound for estimates.